

OLIST BUSINESS DISCOVERY — THE DELIVERY PROMISE GAP

HVIA Data & AI Solutions — Task 01 | Data Analysis Internship

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1. Executive Summary

This report presents a complete business discovery of Olist — Brazil's largest e-commerce enabler — based on 100,000+ real orders (2016–2018) from the public Olist Brazilian E-commerce dataset.

The analysis uncovered a systemic issue in Olist's delivery promise system: it is not failing — it is massively inflated. 78.9% of orders arrive a full week or more BEFORE the promised date, hiding the platform's real delivery speed from its own customers. Meanwhile, the 8.1% of orders that do run late suffer a severe satisfaction penalty: customer ratings collapse from 4.31★ to 2.57★.

Root-cause analysis traced these delays to two measurable sources: slow seller handovers (which multiply customer lateness ×5) and slow shipping routes (which delay orders even with punctual sellers).

The recommended solution — HVIA's Smart Delivery Promise Engine — replaces static delivery estimates with per-order predictions built on seller behavior profiles and route intelligence, improving customer satisfaction and conversion without any logistics restructuring.

2. Company Research Summary

Company Overview

Olist is a Brazilian e-commerce technology company founded in 2015 in Curitiba. It operates as a marketplace enabler: it connects small and medium sellers to major marketplaces and sales channels through a single integration.

Business Model

Olist's revenue comes from SELLERS, not customers:

- Monthly subscription plans
- Commission per sale
- Payment processing fees

Ecosystem (per HVIA brief): ERP management, Marketplace Hub (170+ integrations), Ecommerce storefront, Logistics & Shipping, Payments & Account, Business Credit, AI assistant.

Value Chain

Seller → Olist platform → Marketplaces & Channels → Customer
(Products, Orders, Payments, Shipping, Feedback)

Customers: Small and medium Brazilian sellers who want to sell across multiple channels without managing each marketplace separately.

Key inferred pain points: Maintaining delivery reliability across a country of 8.5M km², keeping thousands of independent sellers accountable, and protecting customer satisfaction (reviews) — the platform's public trust currency.

3. Dataset Understanding

Source: Olist Brazilian E-commerce Public Dataset (Kaggle) — ~100K real orders spanning 2016–2018 across multiple Brazilian marketplaces.

Structure: 9 relational tables imported into MySQL 8.x:

Table	Rows	Role
orders	99,441	Order lifecycle & dates
order_items	112,650	Products per order, price, freight
customers	99,441	Customer identity & location
sellers	3,095	Seller identity & location
order_payments	103,886	Payment methods & values
order_reviews	99,223	Customer satisfaction (1–5)
products	32,951	Product attributes & categories
translation	71	Category PT→EN mapping
geolocation	1,000,163	Zip-level coordinates

Data Quality Findings (documented during validation)

FQ1: 8 "delivered" orders with no delivery timestamp (verified: all 8 carry reviews — 7 rated 5★, 1 rated 1★). A database inconsistency: completed deliveries with no completion time. Excluded from timing analysis; documented.

FQ2: 394 orders carry multiple review rows (edit history preserved as separate rows). Handled via COUNT(DISTINCT order_id); average scores computed over review rows (~0.4% duplication — negligible, documented).

Additional: ~647 delivered orders lack reviews; empty product attributes coerced to 0 during import were converted to NULL; zero dates in orders converted to NULL to protect delay calculations.

Full import challenges (server security restrictions, path formatting, encoding, type coercion, duplicate import) are documented separately in the MySQL Import Challenges Log within this repository.

4. Analysis & Business Story

Insight 1 — The Over-Promised Delivery

Metric	Value
Orders delivered late (after promised date)	8.1%
Orders delivered 7+ days BEFORE promise	78.9%

Olist almost never breaks its promise — but the promise itself is inflated. Nearly 4 out of 5 orders arrive a full week or more early. Every "delivery by day 22" promise for an order that arrives on day 12 represents: a slower-perceived service than reality, and a potential lost sale from any customer who abandons checkout when facing long delivery estimates.

Insight 2 — Late Delivery Destroys Satisfaction

Segment	Orders	Avg Review
Late	7,662	2.57★
Early 1–6 days	12,419	4.18★
Early 7–13 days	33,882	4.30★
Early 14+ days	41,866	4.32★

Total analyzed: 95,829 orders (reconciled against source totals — exclusions documented in Section 3).

Late delivery costs approximately 1.74 rating points (2.57 vs 4.31). Delivery timing — not price, not product category — is a primary driver of customer satisfaction. Notably, satisfaction rises only marginally with extra earliness (4.18 → 4.32), meaning the real damage concentrates in lateness, not in imperfect earliness.

Insight 3 — Where the Delay is Born

Per-state decomposition: seller handover punctuality (carrier pickup date vs. shipping limit date) cross-referenced with customer lateness:

Pattern	Example (seller state)	Reading
Seller-driven	MA: 32.6% seller-late → 24.4% customer-late	Slow sellers cause downstream lateness
Route-driven	ES: 5.5% seller-late → 7.7% customer-late	Carrier/logistics delay despite punctual sellers
Reliable	PE (4.5%→4.5%), GO, RS	Minor issue

Key finding: If the seller hands over late, customer lateness jumps from ~4% to ~22% — a ×5 multiplier. But in certain states, orders run late even with punctual sellers — a pure logistics-path issue.

The Story in One Paragraph

Olist promises delivery dates that are systematically inflated — 79% of orders arrive a week+ early. When delivery genuinely runs late (8.1%), customer ratings collapse from 4.3 to 2.6. The delay is not random: it is born either in slow seller handovers (traceable per seller) or in slow shipping lanes (traceable per route). This means Olist's delivery promise is not a mystery — it is predictable — and today, nobody predicts it. A static, one-size-fits-all estimate is costing Olist on both ends: over-promising hides real speed, and under-predicting lets lateness surprise customers.

5. Solution Proposal — What HVIA Can Offer

Proposed Solution: Smart Delivery Promise Engine

Replace Olist's static delivery estimates with a data-driven prediction layer that computes a realistic delivery date per order at checkout, based on:

1. **Seller behavior profile** — historical handover punctuality (proven: late handover → ×5 customer lateness)
2. **Route intelligence** — origin-destination historical transit performance (proven: some routes are slow despite punctual sellers)
3. **Product/logistics attributes** — weight, dimensions, freight value

HVIA Capability Mapping

HVIA Offering	Role in the solution
Forecasting & Predictive Solutions	Predict delivery date per order at checkout
Logistics Intelligence	Route-level delay detection (state-level lanes)
Analytics & Decision Systems	Seller punctuality scoring & monitoring

Expected Impact

- **Satisfaction:** The 2.57★ late-delivery collapse applies to a shrinking share as promises become accurate.
- **Conversion:** Honest shorter promises ("12 days" instead of "22") reduce checkout abandonment driven by long estimates.
- **Trust:** The 8.1% promise failure rate drops as predictions account for seller and route risk.

Recommended Roadmap for Olist

- **Step 1 (Now):** Audit seller handover punctuality — identify the ×5 risk sellers. The data already exists internally.
- **Step 2 (3 months):** Recalibrate promise dates by route — ES-type lanes need different estimates than SP-type lanes. No single promise fits a country of this size.
- **Step 3 (Long term):** Replace static estimates with the prediction engine — per-order, seller-profile and route-aware.

Each step is measurable with existing data.

6. Outreach Message Draft

Subject: Your delivery promise is 10 days slower than reality — data inside

Hi [Name],

I analyzed Olist's public e-commerce dataset (100K orders, 2016–2018) and found something interesting: 78.9% of delivered orders arrived a week or more BEFORE the promised date — while the 8.1% that ran late collapse customer ratings from 4.3 to 2.6.

The cause is traceable: late seller handovers multiply customer lateness ×5, and specific shipping lanes underperform even with punctual sellers.

At HVIA, we build delivery promise engines that predict realistic dates per order — improving satisfaction and conversion without touching your logistics.

Worth a 15-minute call to walk through the numbers?

Best regards,

7. Technical Appendix

Environment: MySQL 8.x + MySQL Workbench on Windows.

Data pipeline: 9 CSV tables (~1.4M rows) imported via LOAD DATA INFILE with explicit type-safe schemas (DECIMAL for money, VARCHAR for hash IDs, DATETIME for timestamps, NULL-safe fields for incomplete records).

Validation methodology: Every table's row count reconciled against Kaggle reference values; segment sums reconciled after every JOIN; duplicates and truncation caught by post-import COUNT checks.

Reproducibility: Full SQL pipeline (import, validation, analysis) available in this repository's SQL files. Complete error log and solutions available in the MySQL Import Challenges Log document.

Analysis Rules Applied

1. Orders with delivery date → timing analysis only
2. Delivered without delivery date → Data Quality Findings only
3. All totals computed with no filters (source of truth)

After every JOIN → segment sums reconciled against source totals